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OPTIMIZATION OF DENSE PLASMA FOCUS (DPF) NEUTRON SOURCES VIA EXPERIMENTS AND KINETIC MODELING

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Abstract

Intense neutron sources are needed for survivability testing of electronics and other devices. One candidate neutron source is the dense plasma focus (DPF). The DPF is a coaxial plasma rail gun that serves to couple the slow rise time of a capacitor bank (microseconds) with a fast Z-pinch event that accelerates ions into a gas target and produces neutrons in the presence of deuterium or deuterium/tritium gas. The DPF's discharge ends in a 10-100 ns neutron pulse, with total demonstrated yield of up to 10^{12} for deuterium fill gas into 4π , with ~ 1 megajoule of stored energy. DPFs are technologically simple, inexpensive, and compact. Although DPFs have been around for 50 years, until recently these devices have been optimized empirically due to lack of a predictive model. Our group has developed the first fully kinetic model of these devices, in the particle-in-cell (PIC) code, LSP. Through predictive modeling in conjunction with experiments, we seek to increase DPF performance through four techniques: (1) patterned gas loads via a supersonic gas jet, (2) solid targets, (3) a hollow anode for reduced sputter, and (4) the use of UV light to pre-ionize the plasma.

Introduction

The dense plasma focus (DPF) [1], [2] is a candidate intense neutron source for survivability testing, materials testing [3], neutron imaging, radiological source replacement, active interrogation [4], [5], and other applications. DPFs are compact, pulse-power-driven devices consisting of two coaxial electrodes, separated by an insulator and filled with a low-density gas. The discharge of a DPF consists of four distinct phases (Fig. 1). The first phase is the generation of a plasma sheath as the gas flashes over on the insulator. The sheath then "lifts off" the insulator, and begins the second "run down" phase, in which the plasma sheath is accelerated down the electrodes. The third phase is an implosion called the "run in," and the fourth phase is a z-pinch phase in which the plasma stagnates on axis. A given DPF anode is sized relative to the current running through the plasma as well as the expected operating pressure, such that the plasma sheath will arrive on axis simultaneous with maximum current.

During the z-pinch phase, DPFs can produce MeV ion beams, x-rays, and neutrons. The neutrons are produced by interaction of the energetic ion beam with the dense z-pinch target as well as the surrounding background gas, if the gas fill is deuterium or deuterium/tritium. Megaampere-class DPFs with deuterium fills have demonstrated

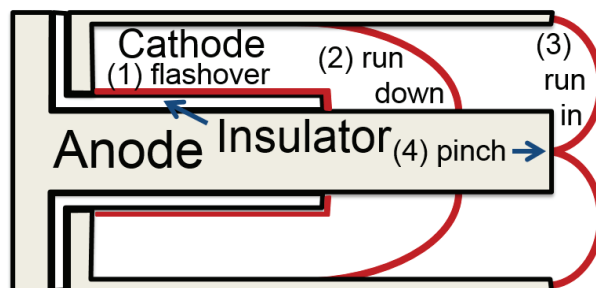


Figure 1. Schematic of a DPF gun. The plasma sheath at different phases of the discharge is represented by the red curves. The majority of the current is carried within the plasma sheath, though parasitic restrikes occurring in trailing mass can divert current from the plasma sheath, reducing the efficacy of the pinch.

neutron yields in the 6×10^{11} [6] to 1×10^{12} (unpublished) neutrons/shot range with pulse durations of 10-100 ns. Due to its short pulse, the DPF is one of the brightest neutron sources in existence and can be made compact: an MJ DPF could be designed to fit on a pick-up truck.

Historically these devices have been empirically optimized. While the device itself is physically simple, the physics behind the ion beam acceleration is complex and not well understood. Past modeling efforts have typically been performed in fluid codes, though it is commonly recognized that fluid approximations break down during the pinch phase of the discharge. The pinch phase features ion and electron beam formation, as well as potentially anomalous resistivity and viscosity. Thus, fluid calculations can predict plasma sheath velocity and timing, but cannot predict neutron yields.

Lawrence Livermore National Laboratory, in conjunction with Voss Scientific, LLC, developed a kinetic (particle-based) model of a dense plasma focus [7], which is now used to optimize DPFs of varying sizes through varying electrode shape, insulator length, operating pressure, and driver properties. The model compares favorably with measurements on the kilojoule [8] and megajoule [9] scale. The kinetic model can also be used to evaluate more exotic non-traditional DPF configurations such as those employing patterned gas loads or solid targets. Here we present a variety of simulation and experimental results aimed at manipulating the dense plasma focus for improved performance.

Kinetic Simulations

Our kinetic simulations are run in the particle-in-cell (PIC) code LSP [10]. These simulations include particle collisions and are 2D axisymmetric. They incorporate electrodes, an external pulsed-power driver circuit, and model the plasma from insulator lift-off through the pinch phase. To accommodate the wide range of relevant spatial and temporal scales within the available computational resources, the simulations are performed using a hybrid fluid-to-kinetic model. This approach allows single simulations to begin in an electron/ion fluid mode from insulator lift-off through run-down of the anode, then transition to a fully kinetic PIC description during the run-in phase, when the current sheath is 3-10 mm from

the central axis of the anode. Simulations are advanced through the final pinch phase using an adaptive variable time-step to capture the femtosecond electron cyclotron motion. Grid spacing is typically 100 μm in the pinch region to resolve relevant plasma features that contribute to ion beam formation and neutron production.

Patterned Gas Loads

In a traditional DPF with a static gas fill, the fill pressure is used to synchronize the pinch event with the maximum plasma current. A higher fill pressure will result in a slower run-down, so that for a fixed anode shape, higher pressure is used when higher charging voltage (and thus more current) is used. Furthermore, most successful DPFs have operated in the 1-10 torr range, presumably because the plasma flashover process is difficult at higher pressures. Being constrained to operate at certain pressures leaves the operator with little control over the imploding mass.

A fast gas puff can be used in a DPF to make a patterned load instead of a static fill, as a mechanism for manipulating the load at the time of implosion while leaving the ambient gas in an acceptable pressure range. A gas valve located at the anode tip can be used to put a supersonic gas jet on axis, which could be used to increase the total yield and to reduce shot-to-shot variation in yield. Super sonic gas flow have been modeled using the large-eddy Navier-Stokes code CharlesX [11]. A sample density profile produced by CharlesX is shown in Fig. 2.

The evolution of the gas profile takes place over several hundred microseconds, while the DPF discharge is at maximum several μs long. Thus, the gas profile is approximately

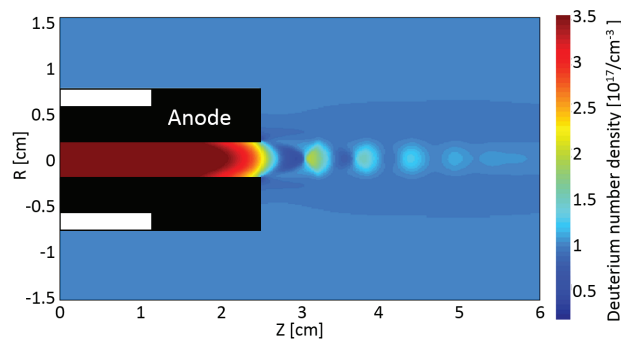


Figure 2. Modeled density profile for a supersonic gas jet into a few torr fill gas, using the CharlesX code. The supersonic flow creates patches of dense gas as well as regions of rarified gas.

static during the discharge. It is possible to port a density profile from CharlesX into LSP for evaluation of the effectiveness of the load.

We use a “miniature” DPF geometry to test out new concepts before running larger simulations. The miniature geometry is small in physical scale, with an anode 1.5 cm diameter, and contains stored energy in the range of 100-300 Joules. Miniature DPF simulations consistently predict that an on-axis density profile with a dip in density improves yields and lowers random yield variability. This pattern can be produced in the laboratory via a Mach diamond structure which forms at the exit of the nozzle in a supersonic gas jet. In past simulations, we have shown

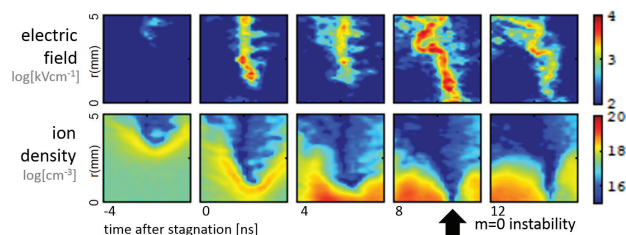


Figure 3. Electric field (top) and plasma ion density (bottom) for 5 subsequent frames in a kinetic simulation that resulted in a strong pinch (high neutron yield). Frames are 5mm x 5mm boxes directly in front of the anode. An $m=0$ instability begins to form in the first frame, and the plasma completely separates in the fourth frame. The electric field fills up the space between the two plasma columns, where the ion beam is accelerated.

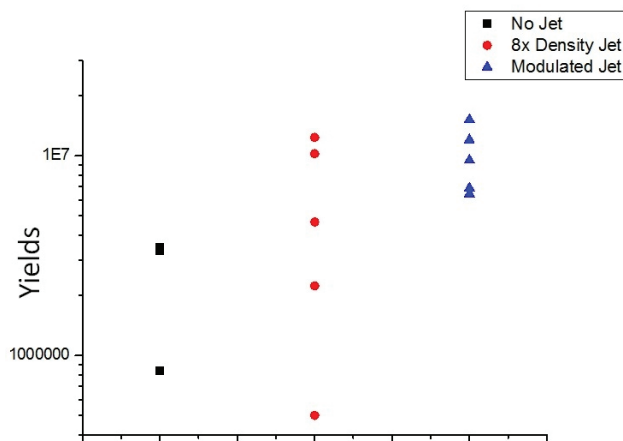


Figure 4. Modeled yields as a function of gas jet configuration. Black squares are from simulations with uniform gas fill (no jet), pink circles are from simulations with a high-density, uniform gas jet on axis, and blue triangles are from simulations with an on-axis gas jet with modulations in it. The modulated gas jet improves yield and reduces yield variability.

that high-yield pinches all have a characteristic strong $m=0$ instability that cuts the plasma in half and creates a cavity on axis where electric field forms (Fig. 3). In the case of a super-sonic jet with Mach diamonds, regions of rarified gas are pre-disposed to pinching earlier than the dense regions, creating a strong $m=0$ instability on the axis. Yields from miniature DPF simulations with a static gas fill, a constant-density gas jet, and a modulated gas jet are shown in Fig. 4.

We have used the Z machine test nozzle to image potential gas loads that can be produced with a fast rise-time gas valve. Spacing between Mach diamonds is controlled by the ratio of pressures in the static fill and plenum. Fig. 5 shows images of 3 gas jets in which the plenum pressure is 40 times the chamber pressure.

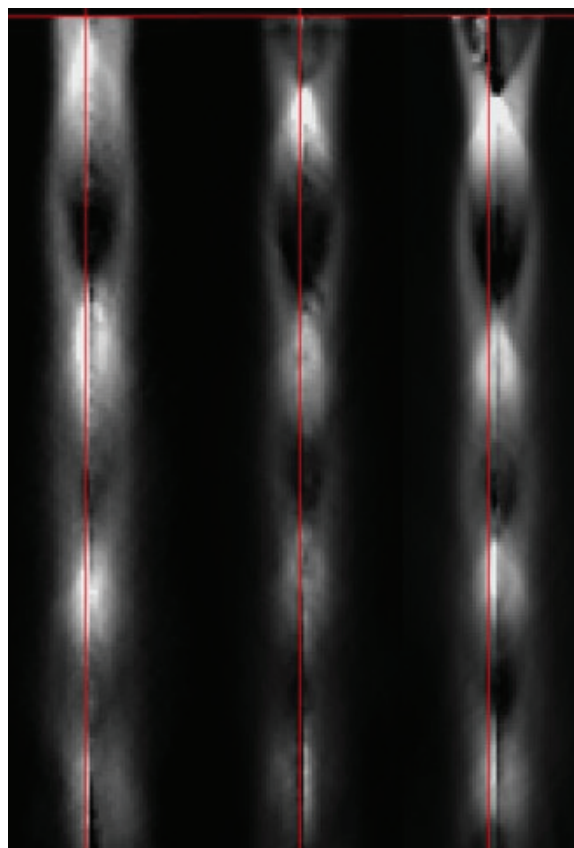


Figure 5. Interferometry images of supersonic gas jet into a static backfill using the Z Machine test nozzle. High-density regions are light and low-density regions are dark. A 40:1 jet-to-chamber pressure ratio is presented in 3 cases: 25-Torr (left), 50-Torr (middle), and 100-Torr (right) backfills. These backfill pressures are on the high side for a DPF chamber fill, but are needed in order to obtain enough signal-to-noise on the interferometer image.

Additional CharlesX modeling of alternative gas nozzle shapes is needed to find configurations that maximize the mass on axis and result in a large density differential (maximum-to-minimum density ratio) in the pinch location. Potential future work includes testing the most promising configurations experimentally.

Solid Targets

Any portion of the DPF ion beam that does not stop in the deuterium gas loses its energy to the vacuum vessel wall. We are investigating potential yield increases from using a solid target downstream of the ion beam. Targets under consideration are deuterated or tritiated titanium, CD_2 plastic, and cryogenic deuterium. Additionally, we have tested and continue to test solid targets on both our smaller kJ DPF at LLNL [12], as well as at MJ-class DPF facilities. Our simulations predict on average a 56% increase in yield using 0.16 steradian CD_2 targets on the kJ DPF. We measured an increase in yield of 53% using this target (Fig. 6).

Yield gains on MJ-class DPFs with solid targets are potentially larger, but it is more challenging to field solid targets in an MJ DPF environment. Our simulations predict a 2.5-5.5x increase in total yield on an MJ DPF with 127 μm thick 3" TiD targets with a 1.6 loading ratio. Our initial experiments on an MJ DPF with this target showed an 82% increase in yield (1.8x), although we discovered that our targets get damaged before they are shot during

warm-up shots with a shutter in front of the target. We have designed a target holder that better protects targets during warm-up shots, to be deployed shortly.

Yield gains using large (several inches diameter) cryogenic deuterium are predicted by the simulations to be much higher for MJ-class DPFs, on the order of 10-100x for thick (several mm) targets. However, producing and protecting such targets in a warm environment is complicated, and we do not currently have any experiments of this type planned. All solid target yield calculations assumed cold matter ion stopping powers.

Reducing Sputter with a Hollow Anode

Motivated by modeling predictions, we performed an experimental scan of anode hollow radius on the LLNL kJ DPF. We found that average yield increased with increasing hollow radius (Fig. 7) at 1.3 torr. Although the reason for the increase in yield is not currently well understood, we do see strong ablation of copper material off of the anode for the flat-faced and smaller hollow anodes, notably at the pinch end of the anode and along the length of the anode (Fig. 8, see next page). This ablation is greatly reduced for anodes with larger hollows and is least present in the anode with the largest hollow. We suspect that the flat-faced anode commonly experiences current restrikes during discharges in which the plasma current gets diverted from the pinch region and is instead flowing in the AK gap. This could lead to ablation on the anode and

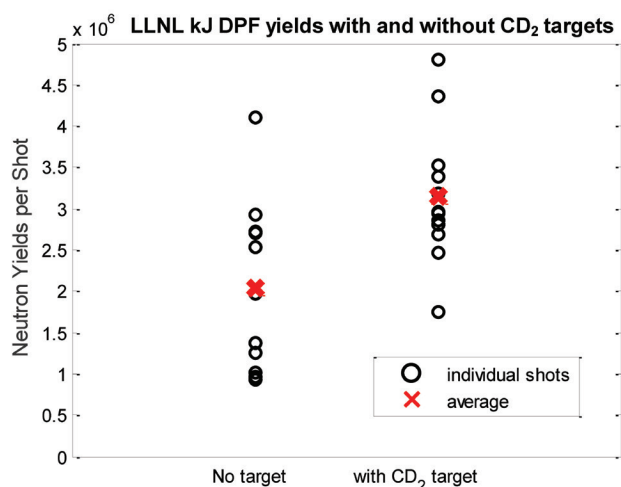


Figure 6. Experimentally measured neutron yields in the LLNL kJ DPF without and with CD_2 targets. Individual shot yields are denoted by circles, while the average is denoted by an "X." The average increase with the target is 53%.

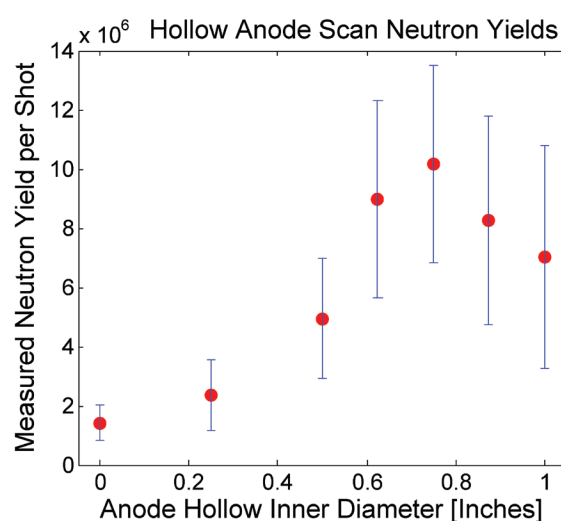


Figure 7. Average yields on the LLNL kJ DPF as a function of anode hollow diameter. Error bars indicate yield variability within 40 shots at the same nominal conditions. All shots are taken at 1.3 torr.



Figure 8. Damage on the side of each anode was most pronounced in the flat anode with no hollow and least apparent in the anode with the largest (1") hollow. We suspect that plasma current restrikes in the AK gap are causing this damage and lead to lower-pinch performance due to lower-pinch current and/or plasma asymmetries.

also contributes to weaker and potentially asymmetric pinch events. The apparent maximum average yield for a 0.75" diameter hollow is within error bars of the other large hollow anodes' yields and may also be an artifact of the constant pressure used throughout the scan since the timing of the pinch changes with each anode due to modified load inductance.

While the hollow anode simulations, performed in a smaller geometry than the experiment, predict an increase in yield with a hollow, they do not predict the yield to increase as hollow radius increases. The model does not include material ablation.

Pre-ionization of the Plasma with UV Light

Typical timing jitter in a traditional DPF neutron pulse can be 100 ns or more, depending on the length of the run-down phase. Some applications require more precise timing than this. We do not currently have a predictive

timing jitter model. The neutron pulse jitter is caused primarily by two phenomena: (1) jitter in the breakdown time of the plasma and (2) differences in shot-to-shot sheath run-down times based on small variance in fill pressure and the presence of impurities. We typically control (2) through a gas purge every shot. In an effort to control (1), we employed a UV diode to assist with plasma breakdown at a specified time. We used a 255 nm diode, which is above the work function of copper, aimed at the cathode plate, to liberate electrons during a 500 ns window. The results shown here are from a 50 mW diode pointed on axis toward the tip of the gun, but we have seen similar effects with a 15 mW diode incident from the side. Overall pinch timing jitter was reduced by approximately a factor of 3 (Fig. 9) by making the breakdown time more consistent. It is somewhat remarkable that an input of energy on the order of nanoJoules can have such a strong effect on breakdown. The use of the UV diode appeared to increase average yield about 10% with both the 15 mW and 50 mW diodes.

Conclusion

A particle-in-cell description of a dense plasma focus Z-pinch allows us to predict the effects of patterned gas loads, the use of solid targets, and electrode shape modification on DPF neutron yields for the first time. We are using this powerful predictive tool to optimize DPFs for

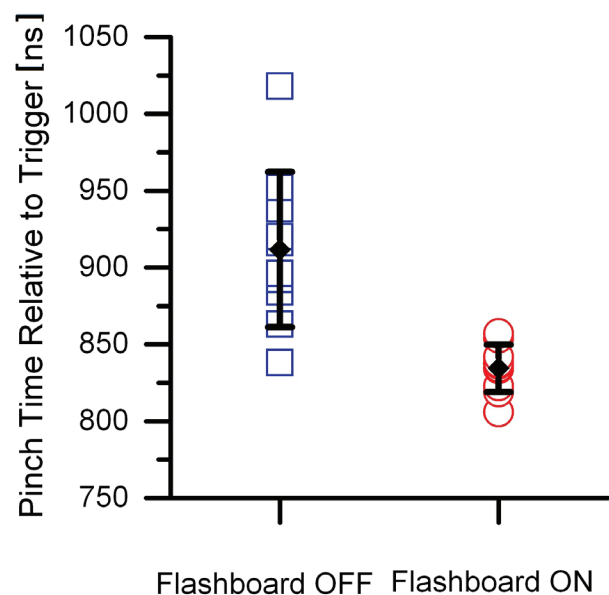


Figure 9. Time of pinch for several shots with the UV flashboard off and the UV flashboard on. The shot-to-shot timing spread is reduced by a factor of ~3 with the UV light.

high neutron yield. Overall, we expect a yield of 10^{13} – 10^{14} per shot on an MJ-class DPF, which may be possible using all of these techniques together, but that may require the use of cryogenic deuterium targets which are non-trivial to produce and shoot.

Additional requirements, such as low-jitter or short-pulse length, are called for in some applications. Here we also present experimental efforts to reduce jitter through pre-ionization with UV light.

Future work with patterned gas loads includes continued gas puff simulations in CharlesX using annular nozzles and other configurations that maximize mass on axis and density differential on axis, and evaluation of these patterned gas loads within LSP. Within the area of solid targets, future work includes fielding such targets on MJ-class DPFs with sufficient target protection during warm-up shots. Additional simulations and experiments are needed to evaluate cylindrical hollow anodes on MJ-class DPFs. We also have plans to test UV pre-ionization on an MJ-class DPF.

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